



GOVERNMENT-READY AGENT RUNTIME

MISSION-READY AGENTS, GOVERNED BY DESIGN.

A customer-hosted, durable agent runtime for regulated and mission environments — every tool call sandboxed, every external action approved by a human, every decision reconstructable after the fact.

PROPOSED SOLUTION

CUSTOMER-HOSTED AGENT RUNTIME WITH AN ENFORCED APPROVAL BOUNDARY

Papyrus is a licensed, customer-hosted daemon for durable, event-driven agent work. Microsoft Entra ID is the sole identity authority, Mastra owns sessions and workflows, and every external side effect crosses a proposal → human-approval → ledger → executor boundary before it happens. Agent code runs sandboxed with network access denied by default. Nothing phones home to a Beag-operated cloud.

THE OUTCOME



OWNED

Customer owns the deployment, the Entra tenant, the license, and the audit trail. No Beag-operated inference in the loop.



BOUNDED

Deterministic policy, roles, and an Entra-authorized approver constrain every action before it reaches the outside world.



DEPLOYABLE

Zarf-packaged for connected, restricted, or fully air-gapped installs, with signed offline licensing and no cloud callback.



AUDITABLE

Integration and action events form an append-only, hash-chained record — evidence that arrives review-ready at first pass.

Prepared for

Federal agencies, prime contractors, national laboratories, and regulated enterprises

General teaming, prototype, pilot, modernization, and production-deployment use

MISSION-READY AGENTS, WITH GOVERNED DECISIONMAKING.

Across the defense industrial base, organizations are being told to adopt AI agents. Almost none of them can do it inside their existing security and governance posture.

THE AGENT ADOPTION GAP

01 VENDOR-CLOUD TOOL SPRAWL

Agent tools and orchestration live in vendor clouds outside the customer's boundary, with no clean way to bring the runtime inside the enclave.

02 UNCONTROLLED DATA EGRESS

Mission and restricted data leaves the enclave to reach hosted models and tool integrations, conflicting with data-control and disconnected-operation requirements.

03 UNACCOUNTABLE ACTIONS

Actions execute without a human authority in the loop, and the resulting decisions can't be reconstructed afterward for audit, incident review, or acquisition evidence.

PAPYRUS RESPONSE

CUSTOMER-HOSTED RUNTIME. ENFORCED HUMAN AUTHORITY. PROVABLE EVIDENCE.

- The entire runtime ships as a hardened, Zarf-packaged container for connected, restricted, or fully air-gapped installs.
- Microsoft Entra application roles are the only identity authority — no local accounts, invitations, or role database inside the product.
- Agent code executes only in a network-denied Bubblewrap sandbox with AgentFS filesystem isolation; any other host refuses execution and reports why.
- Every external side effect crosses a proposal → human-approval → ledger → leased-executor boundary before it happens.
- Integration and action events form an append-only, SHA-256 hash-chained audit trail; licensing is signed, offline, and deployment-bound.

BOUNDARY

Agents propose. They do not approve themselves. Starlings and Mastra generate candidate actions and drafts; a deterministic workflow policy and an Entra-authorized human approver decide what actually leaves the workspace. Consequential outputs are corroborated through validation, review gates, and designated approval — never released on model output alone.

SIX CONTROLS. ONE GOVERNED RUNTIME.

Papyrus runs as a single customer-hosted daemon. Identity, sandboxing, durability, approval, and audit are native to the runtime — not bolted on around a model endpoint.

01

IDENTITY + GOVERNANCE

Microsoft Entra application roles (View, Manage, Security.Manage, Action.Approve, Audit.View, System.Owner) are the only authority. No local accounts, invitations, or role database.

02

SANDBOXED EXECUTION

Agent code runs only on Linux under Bubblewrap with network denied and AgentFS filesystem isolation. Any other host turns itself off and reports why.

03

DURABLE AGENT CORE

Mastra owns durable sessions, memory, schedules, workflows, and signal delivery — independent of any single plugin or adapter.

04

ACTION LEDGER + APPROVAL

Every external side effect crosses proposal → human approval → ledger → leased executor. Starlings proposes; an Entra-authorized approver releases.

05

AUDIT + EVIDENCE

Integration registration, activation, configuration, and disable events form an append-only SHA-256 hash chain, tied to source-code evidence fingerprints.

06

DEPLOYMENT + LICENSING

Zarf packages the runtime for connected, restricted, or fully air-gapped installs. Licensing is signed, offline, and deployment-bound — no vendor cloud callback.

BUILT ON

 ENTRA ID

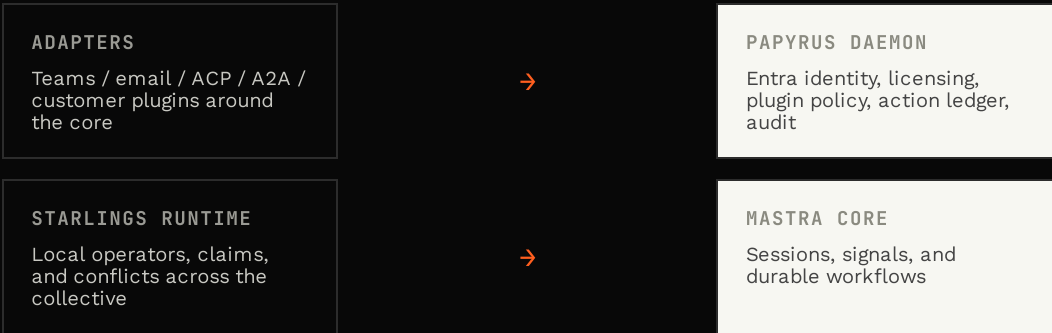
 MASTRA

 BUBBLEWRAP

 AGENTFS

 ZARF

PRODUCT BOUNDARY





THREE BOUNDED WORK PACKAGES.

A focused 60–90 day engagement gives an agency, prime, laboratory, or enterprise team a working, governed Papyrus deployment, delivery evidence, and a clean handoff before either party considers broader scale.

01

RESTRICTED-ENVIRONMENT DEPLOYMENT

Stand up Papyrus inside the customer's Entra tenant and network boundary — connected, restricted, or fully air-gapped.

- Deployment profile and hardware fit assessment (GCC / GCC High / DoD / restricted / disconnected)
- Entra application roles, licensing, and governed plugin lifecycle configuration
- Operating runbook and customer-controlled handoff

02

CONNECTOR + WORKFLOW INTEGRATION

Connect mission systems, SIEM/EDR/OTEL sources, and operational workflows to the agent runtime under approved policy.

- Native connector activation with test → submit → activate lifecycle
- Durable Mastra workflows and scheduled agent tasking
- Webhook and signal-intake configuration scoped to session

03

GOVERNED AGENT PROTOTYPING

Turn a bounded mission requirement into a working, approval-gated agent session inside the program's environment.

- Action-ledger-backed tools, permissions, and human review gates
- Durable session with decision brief and audit evidence
- Demonstration, documentation, and customer handoff

STANDARD INITIAL ENGAGEMENT

FRAME

Confirm users, boundary, Entra tenant, deployment profile, and acceptance criteria.

DELIVER

Implement the bounded capability and exercise it against realistic cases and failures.

VALIDATE

Test identity, sandboxing, approval flow, audit trail, and operator usability.

TRANSFER

Provide runbook, deployment assets, documentation, and customer handoff.

GOVERNED BY DESIGN. ALIGNED TO NIST CSF 2.0 AND THE AI RMF.

Papyrus's identity, sandboxing, approval-ledger, and audit-chain design map directly onto the NIST Cybersecurity Framework 2.0 functions and the NIST AI Risk Management Framework functions. This describes platform capability, not a completed authorization — see the note below.

NIST CSF 2.0

GOVERN Entra-native identity as the sole authority; governed plugin lifecycle (Draft → Tested → Awaiting Approval → Active → Degraded/Disabled); six scoped application roles from View through System.Owner; signed, deployment-bound licensing.
IDENTIFY Live connector and model-gateway inventory (/portal/plugins, /portal/models); typed tool-call contracts with source-evidence fingerprints for continuous asset and dependency awareness.
PROTECT Network-denied Bubblewrap sandboxing for all agent code; AgentFS filesystem isolation; credential-like env vars stripped before execution; secrets accepted only via vault, certificate, or managed-identity reference; Zarf-hardened containers for restricted/air-gapped installs.
DETECT Append-only SHA-256 hash-chained audit of registration, activation, configuration, and disable events; pre-commit and CI drift checks across contracts, workspace, secrets, and generated evidence.
RESPOND Proposal → human-approval → ledger → leased-executor boundary on every external side effect; connectors can be degraded or disabled without disrupting the durable runtime.
RECOVER Durable Mastra sessions, memory, schedules, and workflows persist independent of any single adapter; offline license and model configuration remain intact without a cloud dependency.

NIST AI RMF

GOVERN Human authority is structural, not policy-only; the agent proposes an action and stops; release requires a distinct Entra-authorized Papyrus.Action.Approve decision. Configuration, security-activation, and approval authorities are separately roled.
MAP Customer-controlled model-gateway configuration makes intended use explicit per deployment; deployment profiles (GCC / GCC High / DoD / restricted / disconnected) scope where and how a model is permitted to operate.
MEASURE Structured, typed tool-call contracts tie agent and tool behavior to verifiable, hash-fingerprinted code state; Draft/Tested/Awaiting-Approval status gives each connector a measurable pre-production checkpoint.
MANAGE The action ledger and leased-executor pattern bound what an agent can actually do; abstention is structural — the daemon refuses to chat rather than invent a default model or discard events when none is configured.

IMPORTANT NOTE

This page describes Papyrus's platform design capabilities as they support a customer's own NIST CSF 2.0 and AI RMF programs. It is not a control assessment, certification, or authorization to operate. Papyrus is not, in itself, an authorization to operate, a cross-domain solution, or a claim of FedRAMP, GCC High, DoD IL4/IL6, or SIPR accreditation. Responsibility for risk assessment, System Security Plan content, and AO authorization rests with the deploying customer organization.



DELIVER INSIDE THE PROGRAM.

Beag operates within the customer or prime's program, security, delivery, and governance structure — owning a bounded technical outcome rather than supplying generic staff augmentation.

OPERATING MODEL

■ Program-Aligned	Work is scoped to a funded program, task order, capture, pilot, or customer requirement and managed through the applicable delivery structure.
■ Boundary-Aligned	Access, data handling, model use, tools, repositories, deployment, and evidence follow the program's approved security environment.
■ Outcome-Owned	Beag accepts responsibility for implementation, tests, deployment assets, documentation, demonstration, and customer handoff.
■ Scale-Earned	A successful initial work package creates evidence for progressively larger or more consequential assignments.

COMPANY

ENTITY	PRODUCT	STATUS
Beag Labs, Inc.	Papyrus — government-ready agent runtime	SAM active / Small Disadvantaged Business (SDB)
UEI	CAGE + NAICS	WEB
CDQ1WLG41X6	22QH1 / 541511 / 541512 / 541715	beaglabs.com
CONTACT		
James Bohrman / Founder & CEO		

Desired Next Step

ROUTE PAPYRUS TO THE RIGHT PROGRAM OWNER.

Introduce Papyrus to the relevant capture, technical, program, or mission owner to validate a real requirement and shape a concise 60–90 day work-package concept.